

Supporting Information

Insight into the Formation of Co@Co₂C Catalysts for Direct Synthesis of Higher Alcohols and Olefins from Syngas

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Table S1. Co K-edge EXAFS fitting results for the used Co1Mn/AC catalyst sample after a FTS run treated by syngas firstly at 220 °C, 0.6 MPa for 60 minutes, and then at 250 °C for another 60 minutes.

Sample	Shell	N	R($\text{\AA}$)	$\Delta E_0(\text{eV})$	$\Delta\sigma^2*10^3 (\text{\AA}^2)$	R-factor (%)
Co1Mn/AC-130	Co-C	1.7	1.90	9.5	3.52	0.7
	Co-Co1	4.4	2.53	9.5	8.21	
	Co-Co2	3.4	2.67	9.5	3.52	
	Co-Co3	3.0	2.83	9.5	3.52	
	Co-Co4	1.2	3.02	9.5	3.52	

Table S2. Surface atom ratio of Co^0 to Co^{2+} of the spent $\text{Co}_x\text{Mn}/\text{AC}$ catalysts
calculated by XPS.

Sample	$\text{Co}^0 / \text{Co}^{2+}$
Co1Mn/AC	0.14
Co2Mn/AC	0.10
Co3Mn/AC	0.09

Table S3. Catalytic performance of Co/AC and Co1Mn/AC catalysts before and after passivation in air atmosphere.

	CO conversion (%)	Selectivity (C%)							Alcohol distribution (wt%)				STY (g/kg-cat·h)			Carbon balance (%)
		CH ₄	CO ₂	HC ₂ [±] -HC ₄ ⁼	HC ₂ -HC ₄	HC ₅ [±]	HC ₅ ⁺	ROH	MeOH	C ₂ -C ₅ OH	C ₆ ⁺ OH	C ₂ ⁺ OH	alcohols	olefins	paraffins	
Co/AC	47.5	22.9	0.7	7.0	15.8	7.6	31.3	14.7	10.9	54.0	35.1	89.1	55.6	38.9	204.2	98.4
Passivated	25.4	24.8	1.4	10.3	13.1	10.5	22.6	17.3	15.6	50.5	34.0	84.5	28.0	26.6	74.6	96.1
Co1Mn/AC	29.1	8.1	2.4	17.3	15.5	21.2	14.1	21.4	7.6	55.1	37.3	92.4	46.7	72.3	55.2	99.9
Passivated	28.8	10.0	1.0	17.6	16.7	18.8	15.0	20.9	6.2	55.3	38.5	93.8	39.5	59.9	63.1	97.9

Catalyst tests were conducted at 220 °C, 3.0 MPa, GHSV=2000 h⁻¹ and a H₂/CO ratio of 2.0. Data was obtained and calculated after 12 hours stabilization and 24 h time on stream. After a FTS run, mixed gas of 1% O₂ and 99% Ar was introduced into the reactor to passivate the catalyst, and then the catalyst was exposed to air atmosphere for another 12 hours to achieve a deep passivation. The passivated catalyst was then re-evaluated under the same operating conditions as the previous FTS run.

Table S4. Relative $\text{Co}_2\text{C}/\text{Co}^0$ ratio of the spent Co/AC and Co1Mn/AC catalysts
calculated by XRD and LCF of XANES.

Sample	$\text{Co}_2\text{C}/\text{Co}^0$	
	XRD ^a	XANES ^b
Co/AC	0.6	0.5
Co1Mn/AC	-	3.0

^a Calculated by the peak intensity ratio of Co_2C to Co^0 revised by RIR value.

^b Calculated by the linear combination fitting (LCF) of XANES, relative error 5–10%.

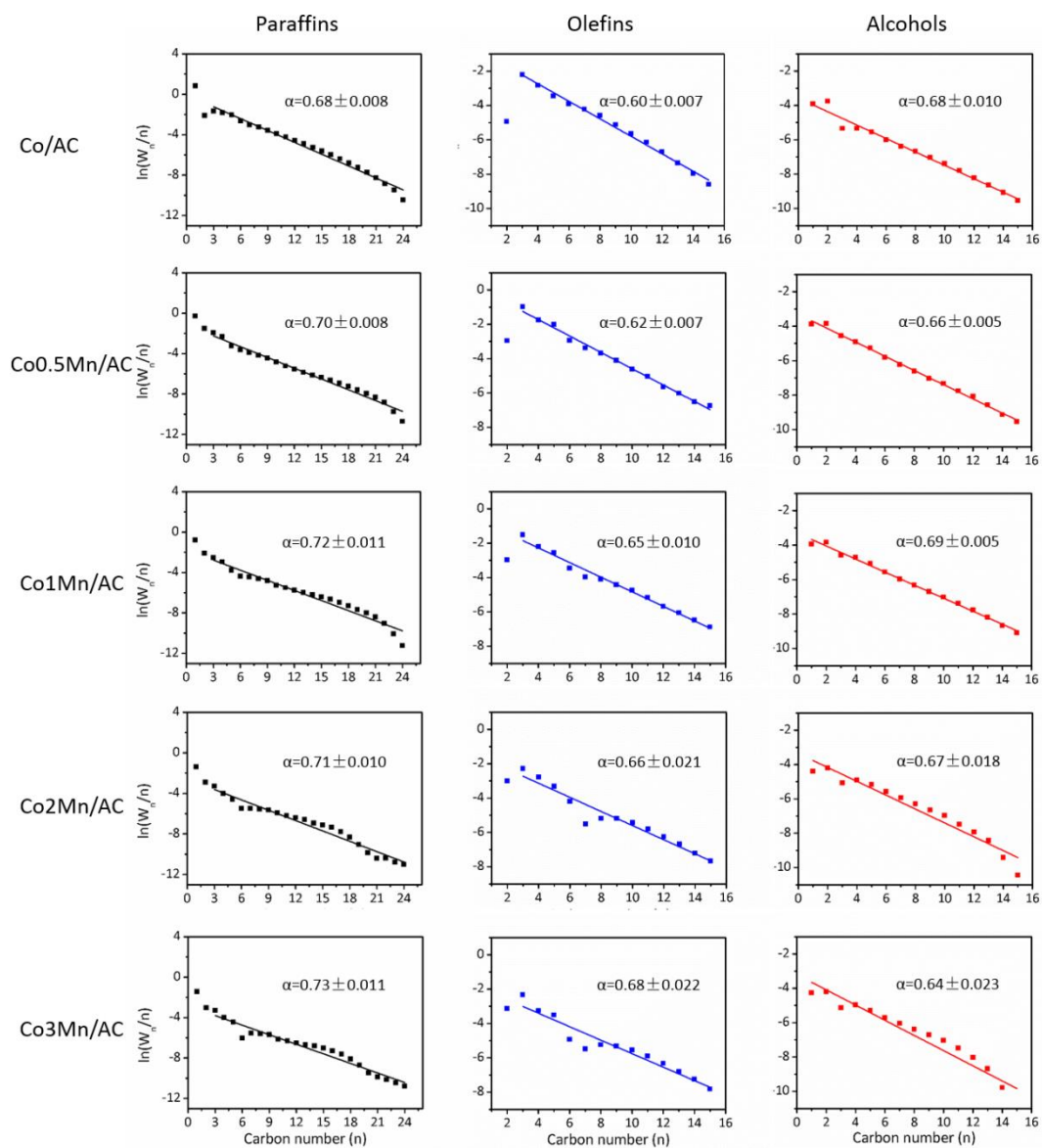


Figure S1. The Anderson-Schulz-Flory distribution of Co/AC catalyst and Co_xMn/AC catalysts.

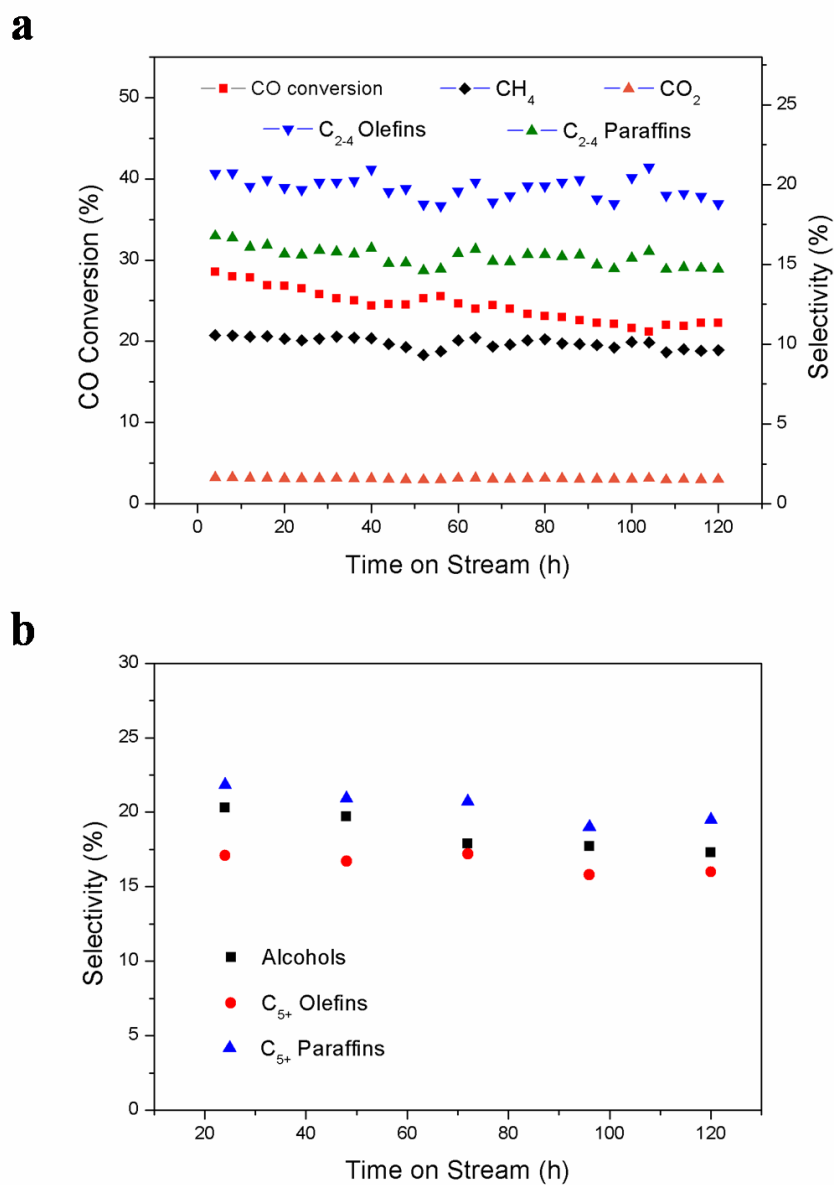


Figure S2. CO conversion and product selectivity over Co1Mn/AC catalyst as a function of time on stream.

Catalytic tests was performed at 220 °C, 3.0 MPa, GHSV=2000 h⁻¹ and a H₂/CO ratio of 2.0.

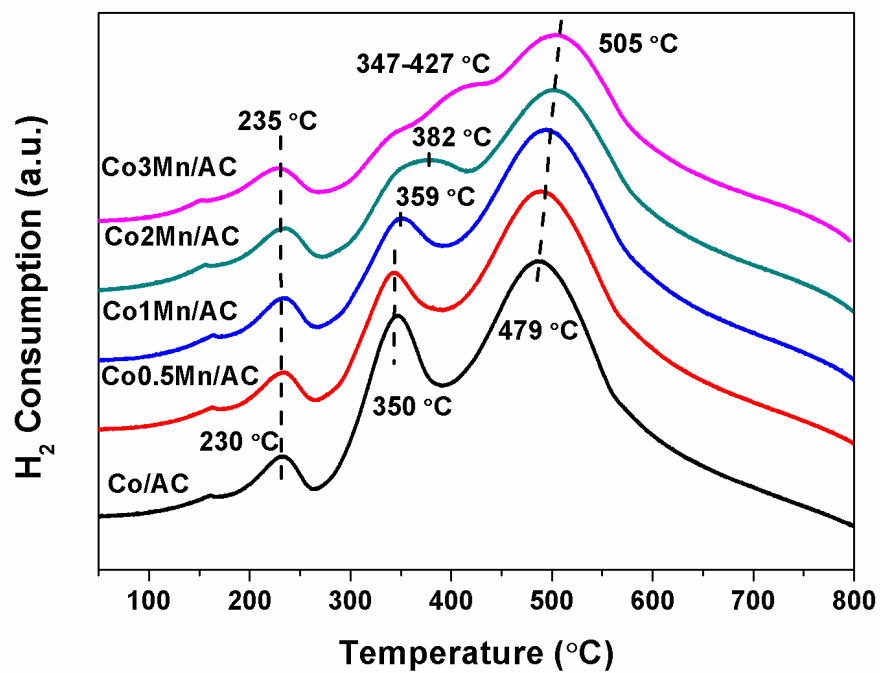


Figure S3. H₂-TPR profiles of Co/AC catalyst and Co_xMn/AC catalysts.

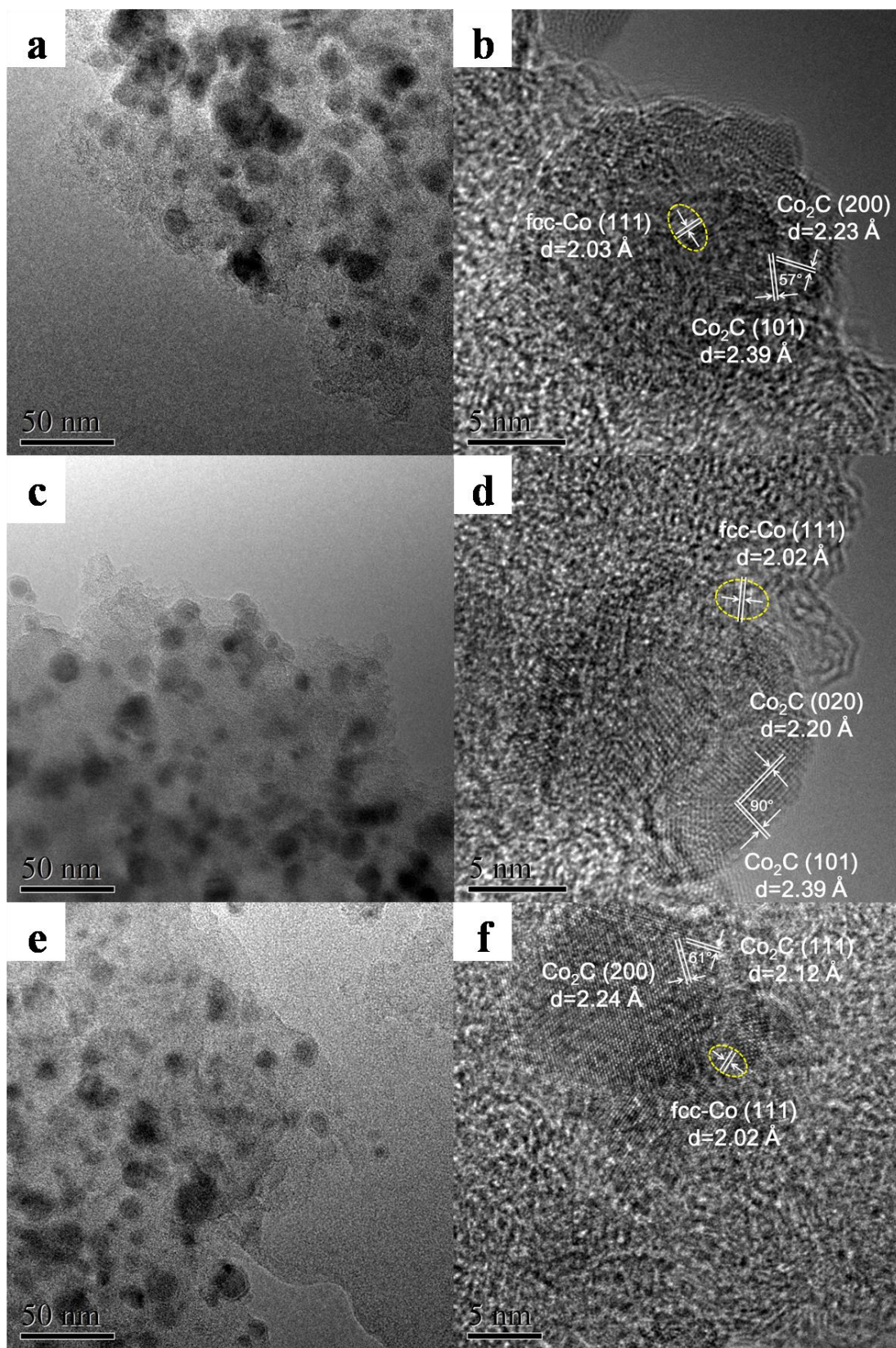


Figure S4. TEM and HRTEM images of spent $\text{Co}_x\text{Mn}/\text{AC}$ catalysts with varying Mn loadings.

(a) and (b) $\text{Co}_{0.5}\text{Mn}/\text{AC}$; (c) and (d) $\text{Co}_2\text{Mn}/\text{AC}$; (e) and (f) $\text{Co}_3\text{Mn}/\text{AC}$.

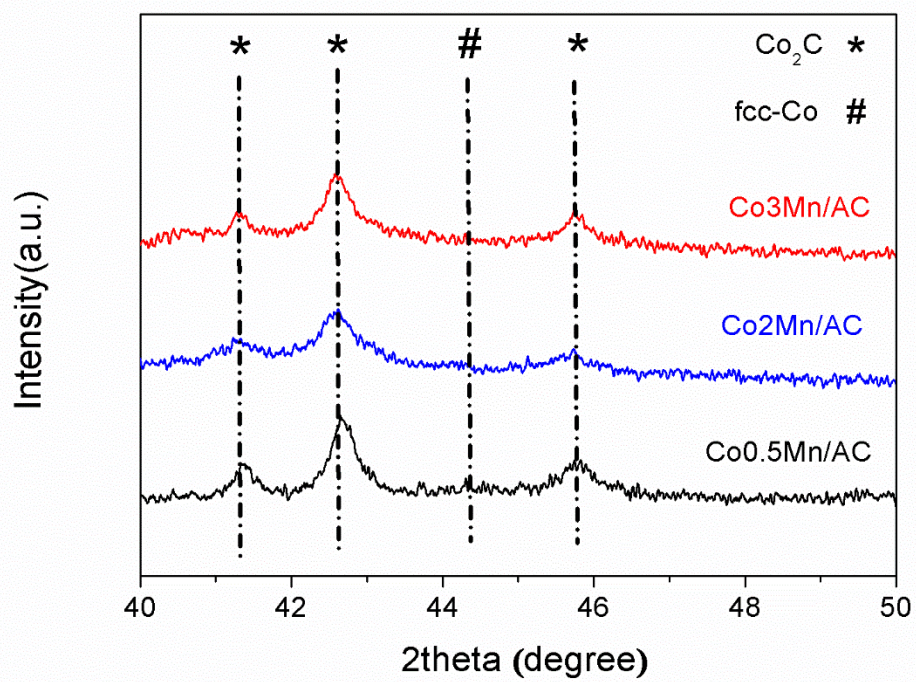


Figure S5. XRD patterns of spent Co_xMn/AC catalysts with varying Mn loadings.

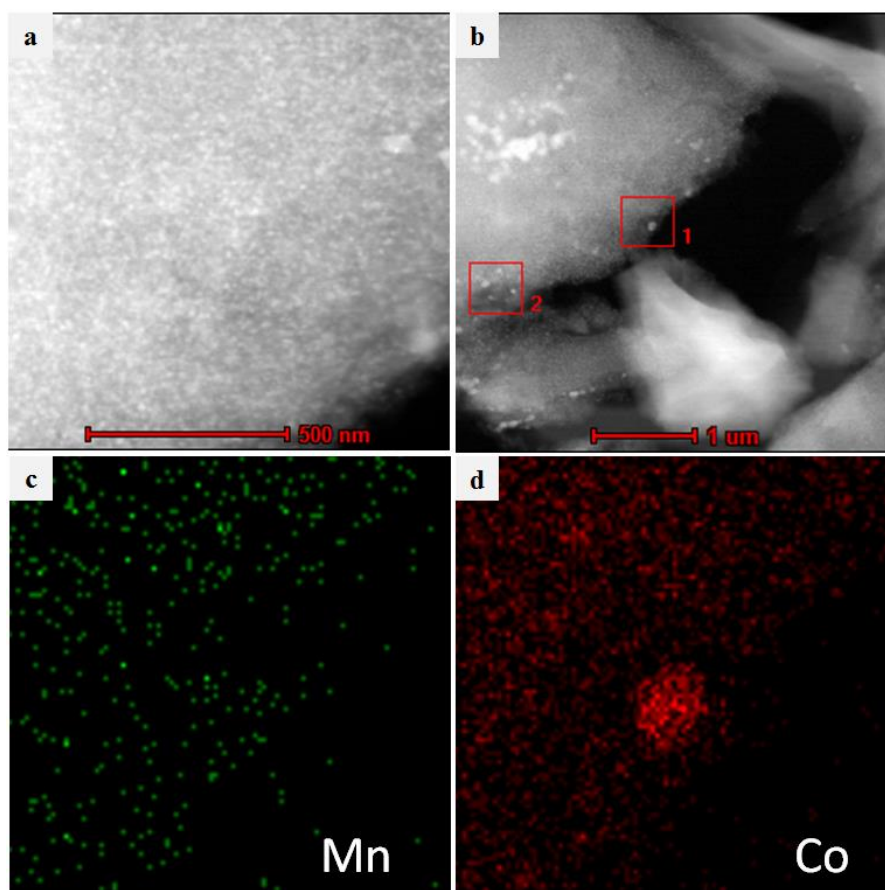


Figure S6. STEM-EDS elemental maps of spent Co₁Mn/AC catalyst.

(a) and (b) STEM image, (c) Mn distribution, (d) Co distribution.

EDS-mapping of catalyst samples prepared by the same procedure were collected by a high-angle annular dark field (HAADF) detector of the Tecnai G2 F30 instrument operated at 300 kV under scanning transmission electron microscopy (STEM) mode.

The *in situ* XRD measurements were conducted PANalytical X'Pert³ Powder diffractometer with Cu K α radiation using an Anton Paar XRK 900 chamber for the *in-situ* treatment of catalyst samples. The phase transformation of the calcined Co1Mn/AC sample was monitored during reduction in a H₂ flow of 20 mL/min with a heating rate of 1 °C/min from room temperature to 430 °C. Each scan was performed every 20 minutes in a 2 θ ranging from 30-70°. Afterwards, the chamber temperature was cooled to 220 °C and catalyst sample was treated by syngas (H₂/CO=2/1) at 0.8 MPa (the maximum pressure of the chamber is less than 1.0 MPa). The XRD patterns were collected every 60 minutes.

A temperature-programmed reduction of calcined Co1Mn/AC sample was performed with *in situ* XRD in H₂ atmosphere under 0.1 MPa. It is seen that fcc-Co starts to appear at 280 °C and the diffraction peak of CoO decreased as temperature was elevated according to Figure S7. The dominating Co species at 400 °C was found to be fcc-Co, showing that reduction of Co oxides was almost complete. Subsequently, no significant change occurred when the operating temperature was elevated from 400 to 430 °C.

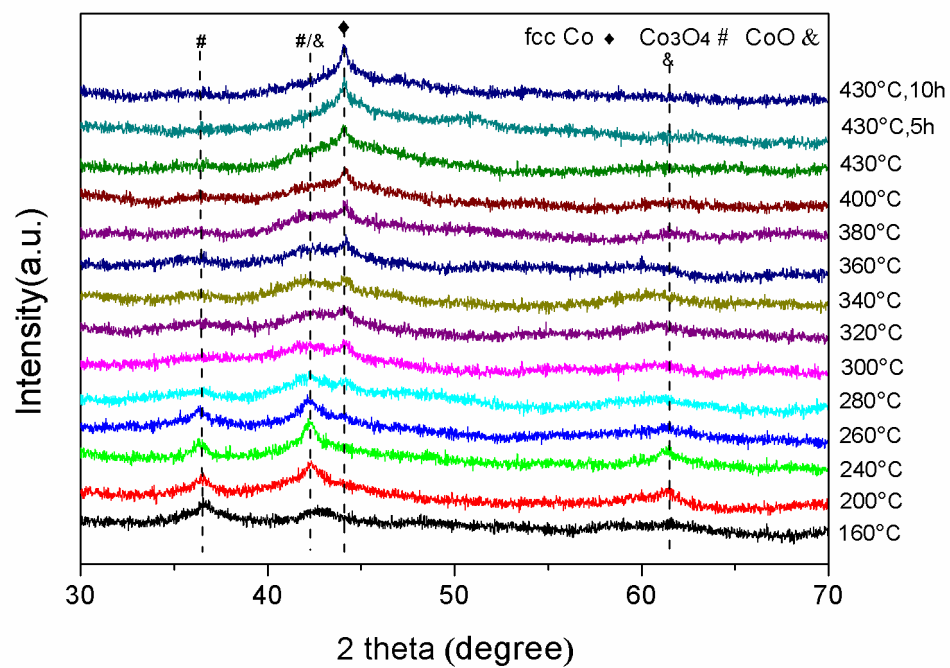


Figure S7. Temperature-programmed reduction of Co₁Mn/AC catalyst in H₂ atmosphere characterized by *in situ* XRD.

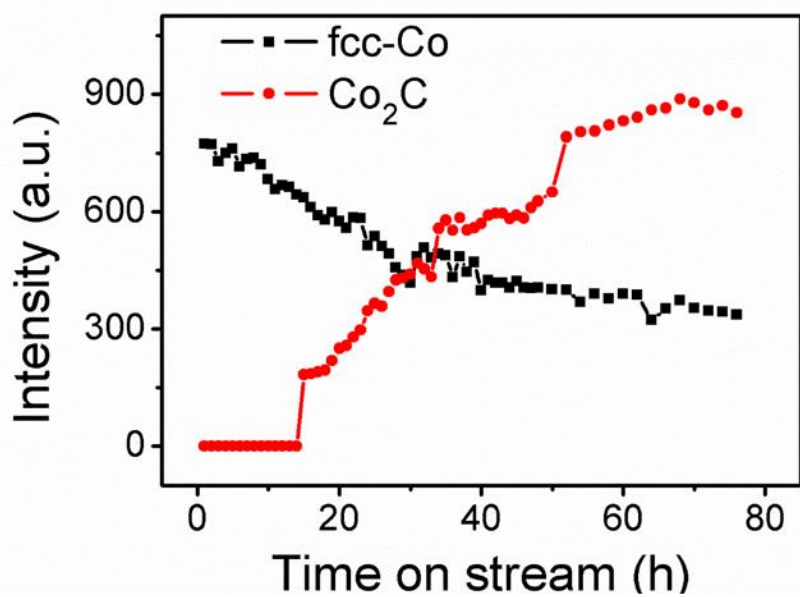
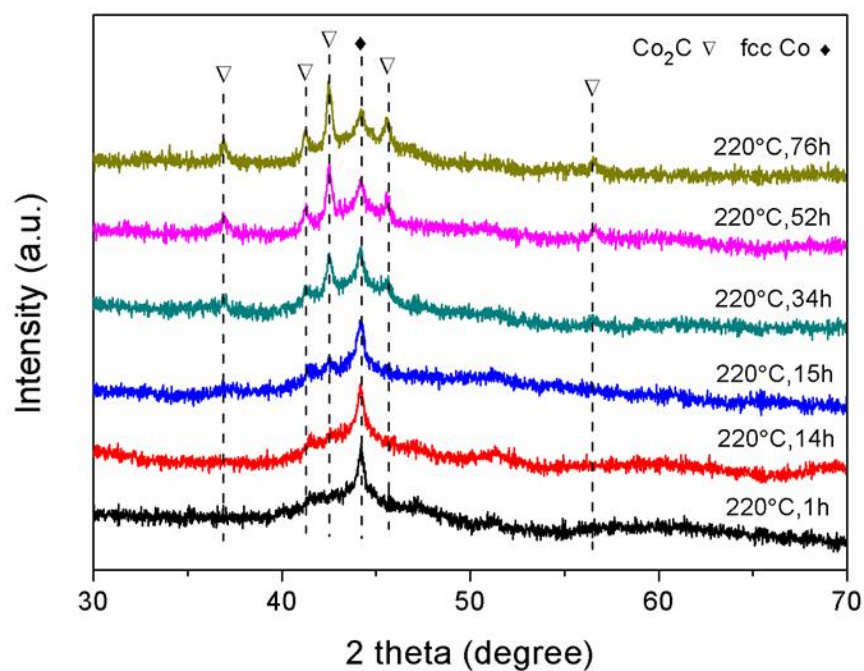


Figure S8. Reaction-induced formation of Co_2C promoted by Mn as a function of time characterized by *in situ* XRD.

Catalyst sample is Co1Mn/AC, and reaction was conducted in syngas atmosphere ($\text{H}_2/\text{CO}=2$), 220 °C and 0.8 MPa.

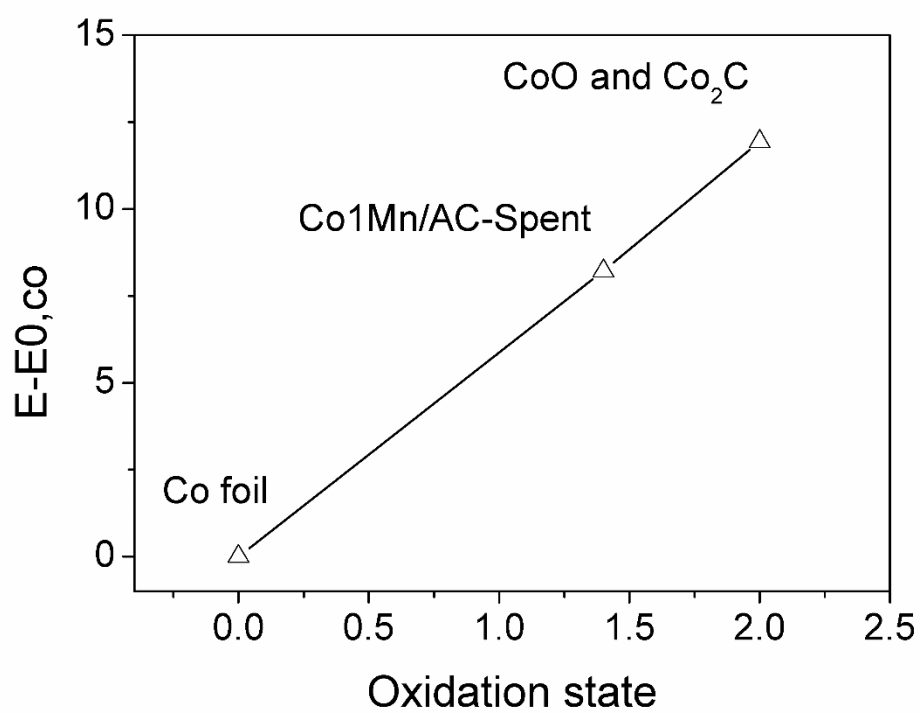


Figure S9. Oxidation state of Co species in spent Co1Mn/AC catalyst, accompanying with reference Co foil and Co_2C .

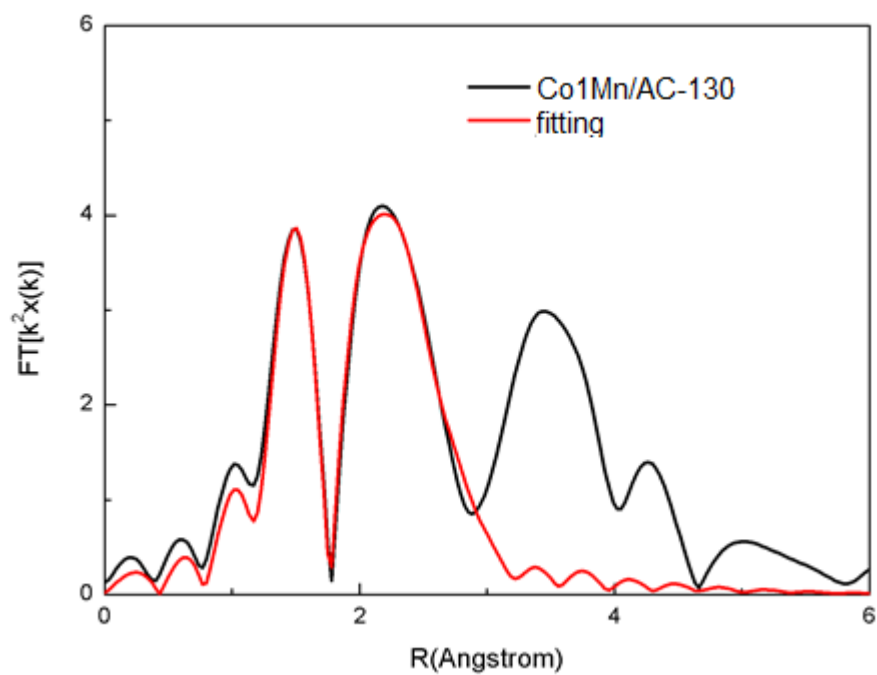


Figure S10. Data fitting of k^2 -weighted EXAFS Fourier transform magnitude spectra of the Co1Mn/AC-130 sample.

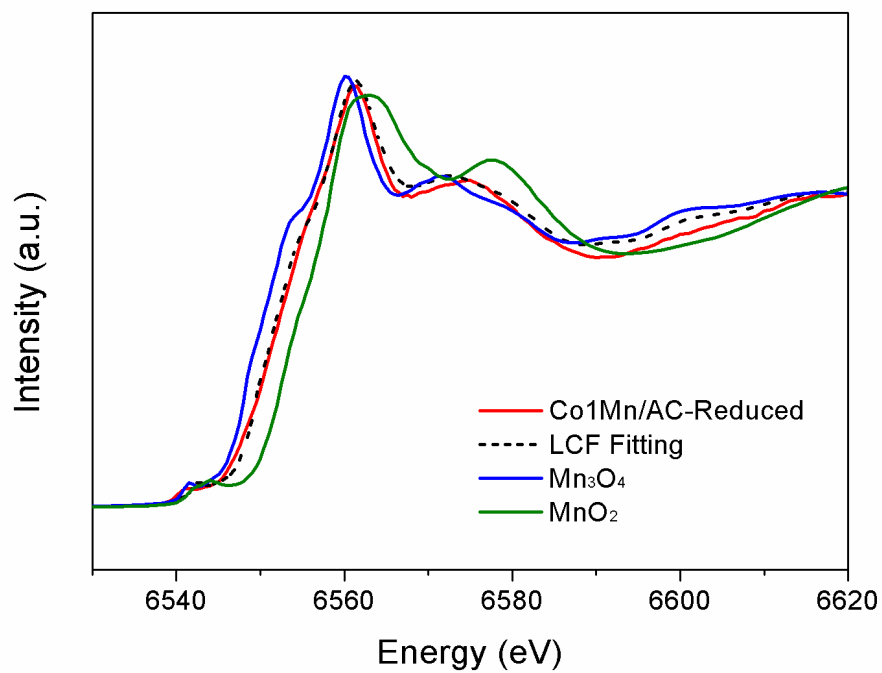


Figure S11. Mn K-edge XANES and its linear combination fitting (LCF) of Co1Mn/AC-Red sample.

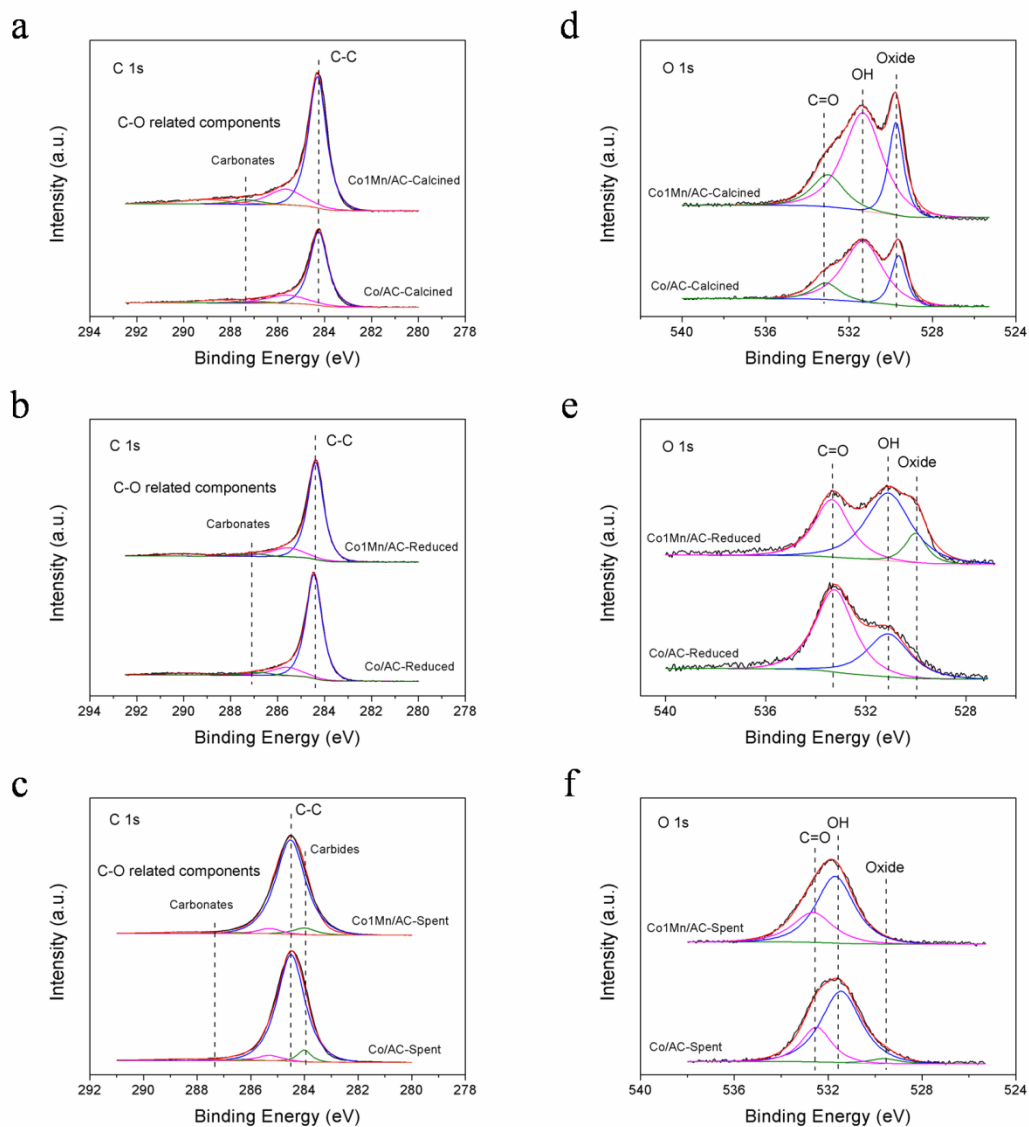


Figure S12. C 1s and O 1s XPS spectra of Co/AC and Co1Mn/AC catalysts.

(a) and (d) calcined at 350 °C in Ar atmosphere for 16 hours; (b) and (e) *in situ* reduced at 430 °C in H₂ atmosphere for 2 hours; (c) and (f) spent after 48 h time on stream at 220 °C, 3.0 MPa, GHSV=2000 h⁻¹ and a H₂/CO ratio of 2.0.